

ASSIGNMENT - 2

#1. Find sum of first 10 natural numbers.

```
sum = 0

for i in range(1, 11):
    sum += i

print("Sum of first 10 natural numbers =", sum)
```

Sum of first 10 natural numbers = 55



#2. Find factorial of a number

```
num = int(input("Enter a number: "))

factorial = 1

for i in range(1, num + 1):
    factorial *= i

print("Factorial of", num, "=", factorial)
```

... Enter a number: 4
Factorial of 4 = 24

#3. Print Fibonacci Series

```
n = int(input("Enter number of terms: "))

a = 0
b = 1

print("Fibonacci Series:")

for i in range(n):
    print(a, end=" ")
    c = a + b
    a = b
    b = c

Enter number of terms: 12
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34 55 89
```

#4. Find largest among 3 numbers

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
num3 = int(input("Enter third number: "))

if num1 >= num2 and num1 >= num3:
    largest = num1
elif num2 >= num1 and num2 >= num3:
    largest = num2
```

```
if num1 >= num2 and num1 >= num3:
    largest = num1
elif num2 >= num1 and num2 >= num3:
    largest = num2
else:
    largest = num3

print("Largest number is =", largest)
```

Enter first number: 10
Enter second number: 20
Enter third number: 30
Largest number is = 30



#5. Student Result System

```
# Input student details
name = input("Enter Student Name: ")
roll_no = input("Enter Roll Number: ")

# Input marks of 5 subjects using loop
total = 0

for i in range(1, 6):
    marks = float(input(f"Enter marks of Subject {i}: "))
    total += marks

# Calculate percentage
percentage = total / 5
```

```
# Display grade using if-elif-else
if percentage >= 90:
    grade = "A+"
elif percentage >= 80:
    grade = "A"
elif percentage >= 70:
    grade = "B"
elif percentage >= 60:
    grade = "C"
elif percentage >= 50:
    grade = "D"
else:
    grade = "F"

# Display Result
print("\n----- STUDENT RESULT -----")
print("Student Name :", name)
print("Roll Number :", roll_no)
print("Total Marks :", total)
print("Percentage :", percentage, "%")
print("Grade :", grade)
```

```
... Enter Student Name: Avni
Enter Roll Number: 40
Enter marks of Subject 1: 30
Enter marks of Subject 2: 98
Enter marks of Subject 3: 91
Enter marks of Subject 4: 60
Enter marks of Subject 5: 89
```

```
----- STUDENT RESULT -----
Student Name : Avni
Roll Number : 40
Total Marks : 368.0
Percentage : 73.6 %
Grade : B
```